PHYSICAL CHEMISTRY CUMULATIVE EXAM

January 16, 1993

Physical chemistry, and science in general, abounds with concepts, rules, equations, and laws which are attributed to their authors/discoverers. Describe/define as many of the following notions as time permits. An equation per se is not sufficient; description of variables and their meaning is also required; plots are good descriptive aids.

(passing score: 12 correct)

Raman effect

Schrödinger equation

Compton effect

Rydberg equation

Stefan-Boltzmann relation

Woodward-Hoffmann rules

Heisenberg uncertainty principle

Beer-Lambert law

Born-Oppenheimer approximation

Hartree-Fock procedure

Morse potential

Walsh diagram

Fermi-Dirac distribution

Wierl equation

Planck distribution

Hooke's law

Coulomb's Law

Bragg law

Miller indices

Ewald sphere

de Broglie relation

Slater orbital

Wien's law

Rayleigh-Jeans law

Monte-Carlo method

Hückel approximation

Larmor frequency

Franck-Condon principle